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COMPLETE NOTES

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Chapter: 5 Introduction to probability

The theory of prob. has its origin in the games of chance related to gambling for instance, throwing of dice or coin, drawing cards from a pack of cards and soon. Gerolamo Cardan (1501-1576) an Italian mathematician was the first man to write a book on the subject entitled "Book on Games of Chance" which was published after his death in 1663. It is a valuable treatise on the hazards of the game of chance and contains a number of rules by which the risks of gambling be minimized and one could protect oneself against cheating.

फ्रेंच, French pascal (1623-62) and pierre de fermat ने यह Scientific development of the subject of probability.

- Swiss mathematician James Bernoulli (1654-1705) → "Treatise on probability."

...
Research से ज्ञान

Theory of prob. → (Foundations of prob.)

- The prob. is the chance or possibility or likelihood of occurring or not occurring on event under an experiment having the numerical value from 0 to 1.
- If prob. is 0, then it represents the impossible event. Similarly if prob. is 1, then it represents the Sure event.

Initially, the application of prob. was made by European mathematicians.

mathematically it can be expressed as

$$P(A) = \frac{M}{N}$$



where,

M = Total no. of favorable event

N = ~~Total no. of possible event~~

$P(A)$ = Prob. of event A

Basic Terminology

- Experiment → Science का Lab मा गर्ने experiment.
↳ करिव-करिव result सट्टै हुन्छ ।

- Random Experiment:

If an experiment performed a large number of times under essentially an identical condition, the result may not be unique but may be any one of the various possible outcomes is called random experiment. Tossing an unbiased coin and rolling a uniform dice are same example of random experiment.

✓ Homogeneous condition मा result फरक-फरक आउछ ।



- Trial and Event:

→ experiment performance गर्नु trial है
गने result आउनु event है ।

Performing a random experiment is a trial.
That outcomes or a set of outcomes of an activity is called an event. Thus an event
be define as the Subject of Sample

space. Just for example for flipping an unbiased coin is a trial and getting either head or tail is an event.

Sample Space: Total set of possible outcomes. It is denoted by (S) .

$$S = \{H, T\}, \quad S = \{1, 2, \dots, 6\}$$

Types of event:



• Mutually exclusive event:

- आफू आउदा अरुलाई आउन नदिने
happening of one precludes happening of another. जस्तै coin मा H आउदा T आउँदैन

• Equally likely events / case:

- आउने chance सबैको बराबर भएमा त्यसलाई equally likely event भनिन्छ।
- occurrence of all possible event are same.

(पहाडवाट कुनै व्यक्तिले पारासुट नलिइएर हामफाल्यो भने त्यो चाही equally likely हुने)

Events or cases are said to be equally likely if there is no reason to expect one in preference to other. In tossing an unbiased coin, head and tail are equally likely.

✓ Exhaustive cases:

case are said to be exhaustive when it includes all possible outcomes of a random experiment. In tossing a coin one the exhaustive no. of cases are head 'H' and Tail 'T'. Therefore the exhaustive no. of cases = 2

Similarly,

In rolling a six faced dice exhaustive number of cases = 6.



Dependent and Independent:

यदि coin tossed गर्दा पहिला (H) आयो तर त्यही coin फेरि tossed गर्दा यदि रुक अर्थात् पहिलो experiment ले 2nd experiment लाई कुनै असर पर्दैन भने यसलाई Independent भनिन्छ। ~~with~~ replacement को case independent हुन्छ। यदि पहिलो experiment ले दोस्रो experiment लाई असर पर्ने गर्छ भने त्यो dependent हुन्छ। with replacement को case dependent case हो।

Events are said to be independent if the happening of one event does not affect the happening of the other event. In tossing of a coin the occurrence of head in first tossing is independent if the occurrence of head in the second tossing

Events are said to be dependent if the occurrence of one event affects the prob. of the occurrence of the other event. Let us consider an example of drawing cards. Let us draw a king card from the pack of 52 playing cards.



Favourable Cases or Events:

The number of outcomes of a random experiment which results in the happening of an event is termed as the cases favourable to the event. For example: in drawing a cards from a pack of 52 playing cards

the number of cases favourable to drawing a queen is 4.

classical / prior / mathematical approach

$$\rightarrow P(E) = \frac{n}{N} \rightarrow \frac{\text{favourable case}}{\text{Total no. of cases}}$$

যৌ equaly likely & mutually exclusive event সা সাত্র লাগু হুন্ডে।

Range $\rightarrow (0-1)$ সা হুন্ডে।

0 $\rightarrow$ Null event (Impossible)

1 $\rightarrow$ Certain event (fixed)

নস্রাডনে (non. occurrence) $P(\bar{E}) = 1 - P(E)$



$$\therefore P(E) + P(\bar{E}) = 1$$

Relative frequency / Statistical
relative frequency approach

যৌ passed experiment সা depernt হুন্ডে।
Experiment গরুপে। यसस्र number of trial
थेरै हुन्डे। यौ कठि व्यापार व्यकसायसा प्रयोग
हुन्डे।

Subjective approach अनैन्नी चानी
आफ्नो individual apperance फिनु

✓ ✓ ✓



~~Permitted~~

Permutations

$$n_p = \frac{n!}{(n-r)!}$$

Combination

$$n_c = \frac{n!}{r!(n-r)!}$$

9. A committee of four members is formed out of 5 statisticians, 7 engineers, 6 economists and 4 doctors. find the prob. that the committee includes.

- (a) Consists one of each kind (b) all of them from same profession (c) 2 statisticians and 2 doctors (d) 1 doctor and rest other (e) at least one statistician.

Solⁿ,

Here,

Given that

Statisticians 5

Engineers 7

Economists 6

Doctors 4

Total

22 $\Rightarrow$ 4 member should be randomly picked.

we have,

$$\begin{aligned} \text{Total possible case (N)} &= 4 \text{ from all 22 members} \\ &= \frac{22!}{4!(22-4)!} \\ &= 7315 \end{aligned}$$

Req. (a) one of each kind

favorable case (M) = one from each kind

= 1 from statistician, 1 from Engineer, 1 from Economist and 1 from Doctors.

$$\begin{aligned} &= {}^5C_1 \times {}^7C_1 \times {}^6C_1 \times {}^4C_1 \\ &= 280 \end{aligned}$$

$$\therefore \text{req. prob.} = \frac{M}{N} = \frac{840}{7315} = 0.115$$

Req. (b) All from same profession.

Favourable case (M) = All from same profession
 = Either all 4 from statistician
 or all 4 from Engineer or
 all 4 from Economist or all
 4 from doctor



$$= {}^5C_4 + {}^7C_4 + {}^6C_4 + {}^4C_4$$

$$= 5 + 35 + 15 + 1$$

$$= 56$$

$$\therefore \text{req. prob.} = \frac{M}{N} \Rightarrow \frac{56}{7315}$$

$$\Rightarrow 0.0076$$

Req. no. (c) 2 statisticians and 2 doctors

Favourable case (M) = 2 statisticians and 2 doctors.



= 2 from statistician and
 2 from doctor

$$= {}^5C_2 \times {}^4C_2$$

$$= 10 \times 6$$

$$= 60$$

$$\therefore \text{Req. prob} = \frac{M}{N} \Rightarrow \frac{60}{7315} = 0.0082$$

Req. no. (d), 1 doctors and rest other

fav case (M) : 1 doctor and 3 from
Non-Doctors

$$= {}^4C_1 \times {}^{18}C_3$$

$$= 4 \times 816$$

$$= 3264$$



$$\therefore \text{Req. prob.} = \frac{M}{N}$$

$$= \frac{3264}{7315} \Rightarrow 0.4462$$

req. (e) At least 1 statisticians

we know,

$$P(\text{At least statistician}) = 1 - P(\text{Non-statistician})$$

Finding favorable case for None is statistician

fav. case (M) = None is statistician

= All 4 from non-statistician

$$= {}^{17}C_4$$

$$= 2380$$

$$P(\text{Non-statistician}) = \frac{M}{N} = \frac{2380}{7315} \Rightarrow 0.325$$

$$\therefore \text{Req. probability} = 1 - 0.325 \\ = 0.675$$

✓

Probability of Events in favor and Against

If odds in favour of an event are $a:b$ then the prob. of A is obtained as: $P(A) = \frac{a}{a+b}$

Similarly,

If odds against of an event are $a:b$ then the prob. of A is obtained as:

$$P(A) = \frac{b}{a+b}$$

Where,

a = front value of ratio (i.e. success)

b = Back value of ratio (i.e. failure)

The odds against A speaking the truth are $1:3$ while the odds in favour of B speaking the truth are $5:4$. What is the prob. that A and B contradict each other in stating the same fact?

Solⁿ

Here,

Let A = Event of A speaking truth.

B = Event of B speaking truth.

Then we have,

$$P(A) = \frac{3}{4+3} = \frac{3}{7}$$

$$P(\bar{A}) = 1 - \frac{3}{7} = \frac{4}{7}$$

$$P(B) = \frac{5}{5+4} = \frac{5}{9}$$

$$P(\bar{B}) = 1 - \frac{5}{9} = \frac{4}{9}$$

Now,

$P(\text{Both contradict each other}) = P(A) \times P(\bar{B})$ or

$$P(B) \times P(\bar{A})$$

$$= \frac{3}{7} \times \frac{4}{9} + \frac{5}{9} \times \frac{4}{7}$$

$$= \frac{32}{63}$$

Theorems of probability

यदि prob. विवर prob. में सौ. दीने आवश्यकता सांख्यिक theorems लगाउनुपर्छ।

- The theorems of prob. must be applied at that condition when the probability of happening or not happening events are already given in the question.
- There are two theorems of prob. and they are given below:

1. Multiplicative Theorem.
2. Additive Theorem.

Multiplicative Theorem

- The prob. of two or more independent events occurring together or in succession is the prob. of their respective probabilities.

for example:

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B)$$

$$P(A, \text{ and } B \text{ and } C) = P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$$

Note: If the terms both, all & and are given, then in that case we use multiplicative theorem.

Additive Theorem

- The prob. of occurring at least one event from two or more events is the sum of their respective probabilities.
- There are two condition for applying additive theorem.

1. For mutually Exclusive Events:

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$$

2. For mutually inclusive Events:

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Note: If the terms like at least, one of them & Either - or are given, then in that case we use additive theorems.

A card is drawn from a well shuffled pack. what is the prob. that the card will be a king or a queen?

Solution

Let

A = event of drawing a king card.

B = event of drawing a queen card.

Then,

$$P(A) = \frac{4}{52}$$

$$P(B) = \frac{4}{52}$$

Hence the prob. of drawing either a king or a queen in a single draw is,

$$P(A \text{ or } B) = P(A \cup B)$$

$$= P(A) + P(B)$$

$$= \frac{4}{52} + \frac{4}{52}$$

$$= \frac{2}{13}$$

8. A card is drawn from a well shuffled pack. what is the prob. that the card will be a king or a Spade?

Solⁿ
Here,

Let,

A = event of drawing a king card

B = event of drawing a space card.

Then,

$$P(A) = \frac{4}{52}$$

$$P(B) = \frac{13}{52}$$



Since, there is a common card for king of Spade, then

$$P(A \cap B) = \frac{1}{52}$$

Hence, the prob. of drawing either a king or a Spade in a single draw is,

$$\begin{aligned} P(A \text{ or } B) &= P(A \cup B) \\ &= P(A) + P(B) - P(A \cap B) \\ &= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} \\ &= \frac{16}{52} \\ &= \frac{4}{13} \end{aligned}$$

9. The prob. that a employees travel from bus is 0.80 and the prob. that they travel from bus and taxi 0.40. The prob that they use at least one means of vehicle is 0.60.

(a) Find the prob. that they travel from taxi.

(b) Are both events statistically independent?

Solution

Here,

Let A = event of traveling from bus

B = event of traveling from taxi

Then,

$$P(A) = 0.80$$

$$P(A \cap B) = 0.40$$

$$P(A \cup B) = 0.60$$



Req. (Q) $P(B) = ?$

We know,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{or, } 0.60 = 0.80 + P(B) - 0.40$$

$$\text{or, } 0.60 - 0.80 + 0.40 = P(B)$$

$$\therefore P(B) = 0.20.$$

Req. (R) Testing Independence

$$P(A \cap B) = P(A) \times P(B)$$

$$= 0.80 \times 0.20$$

$$= 0.16$$

Decision:

Since $P(A) \times P(B) \neq P(A \cap B)$, the given events are not statistically independent.

The probability of three students solving a problem of statistics are in the ratio 2:3:4. Find the prob. that the given problem will be solved.

Soln

Here,

Ratio for solving the problem by 3
Students = 2:3:4

Total Ratio $\Sigma R = 2+3+4 = 9$

Let, A, B and C be the events of solving
the problem by 3 students respectively.
Then,

$P(A) = 2/9$  $\therefore P(\bar{A}) = 1 - 2/9 \Rightarrow 7/9$

$P(B) = 3/9 = 1/3$ $\therefore P(\bar{B}) = 1 - 1/3 \Rightarrow 2/3$

$P(C) = 4/9$ $\therefore P(\bar{C}) = 1 - 4/9 \Rightarrow 5/9$

$P(\text{problem will be solved}) = ?$

As per the information given in the
question, the prob. of problem will be
solved can be expressed as the prob.
that at least one can solve the problem.
Then we get,

$P(\text{problem solved}) = P(\text{At least 1 can solve})$

$= 1 - P(\text{none can solve})$

$= 1 - [P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})]$

$= 1 - [7/9 \times 2/3 \times 5/9]$

$= 1 - 0.096$

$= 0.904$



Conditional Probability

The conditional prob. of an event occurred at that time when another event is already occurred.

It means, under the conditional prob. the occurrence of one event is influenced (i.e. affected) by the occurrence of another event.

Conditional prob. is the prob. of an event with reduced sample space.

Conditional prob. is the prob. of dependent events.

It is calculated as,



$$P(A/B) = \text{prob. of occurring event A when event B is already occurred} \\ = \frac{P(A \cap B)}{P(B)}$$

$$P(B/A) = \text{prob. of occurring event B when event A is already occurred}$$

$$= \frac{P(A \cap B)}{P(A)}$$

Some facts about conditional probability

The conditional prob. is valid only when $P(A)$ and $P(B)$ are greater than zero.

Under the question of conditional prob., two statements are always given.

In one statement, base is given with past tense or present perfect tense. Sometimes, base statement is expressed with 'if' clause.

or "when clause"

• Similarly in another statement, we are given. Asked event with present tense or future tense.

• A bag contains 8 red and 6 white balls. Two balls are drawn randomly from the bag one after without replacement, find the prob. that both balls are white.



Solⁿ

Here,

Prob. of drawing a white ball in the first attempt = $P(A) = \frac{6}{14}$

Prob. of drawing a white ball in the second attempt given the condition that one white ball is already drawn = $P(B/A) = \frac{5}{13}$

Required prob. of drawing both white balls,

$$P(A \cap B) = P(A) \cdot P(B/A) \Rightarrow \frac{6}{14} \times \frac{5}{13}$$

$$\Rightarrow 0.1648.$$

• A bag contains 8 red & 5 white balls. Two successive drawings of 3 balls are made such that (a) balls are replaced before the second trial, (b) the balls are not replaced before second trial. Find the prob. that the first drawing will give 3 white and the second red balls.

Solⁿ

Here,

Let, A denotes the event of drawing 3 white balls in the first draw and B denotes the event of drawing 3 red balls in the second draw.

a) with replacement case: $P(A) = \frac{{}^5C_3}{{}^{13}C_3}$

$$P(B) = \frac{{}^8C_3}{{}^{13}C_3}$$

Required prob. i.e. $P(A \cap B) = P(A) \times P(B)$



$$= \frac{{}^5C_3}{{}^{13}C_3} \times \frac{{}^8C_3}{{}^{12}C_3} \Rightarrow \frac{10}{286} \times \frac{56}{286} \Rightarrow 0.007$$

b) without replacement case:

$$P(A) = \frac{{}^5C_3}{{}^{13}C_3}, P(B|A) = \frac{{}^8C_3}{{}^{10}C_3}$$

$\therefore$ req. prob. i.e. $P(A \cap B) = P(A) \times P(B|A)$

$$= \frac{{}^5C_3}{{}^{12}C_3} \times \frac{{}^8C_3}{{}^{10}C_3}$$

$$= \frac{10}{286} \times \frac{56}{120}$$

$$= 0.016.$$

Nepal Agriculture Bank has records which show the following analysis of its two officers.

Age	Bachelor's degree.	Master's degree.	Total
Under 30	40	10	50
30 to 40	20	30	50
Over 40	90	10	100

If one officer is selected at random from bank. find

- (i) The prob. that he has a bachelor's degree.
- (ii) The prob. that he has master's degree given that he is over 40.
- (iii) The prob. that he is under 30 given that he only a bachelor's degree.

Solⁿ

Here,

Let A be the event that an officer has only a bachelor's degree, B is that he is over 40 years of age and C is that he has master's degree and $\bar{B}$ is that he is under 30 years of age.

- (i) The prob. that he has only a bachelor's degree is

$$P(A) = \frac{150}{200} \Rightarrow \frac{3}{4}$$

- (ii) The prob. that an officer is under 30, given that he has only a bachelor's degree is

$$P(\bar{B}/A) = \frac{P(\bar{B} \cap A)}{P(A)} = \frac{40/200}{150/200} = \frac{4}{15}$$

✓

Q. A Company has made available to its employees (without charge) extensive health club facilities may be used before work, during lunch hours, work, and on week ends. Records for the last indicates that of 250 employees, 110 used the facilities at least once.

the company, 65 used the facilities

Set up the (2×2) table to evaluate the probabilities.

what is the prob. that an employee chosen at random is a female and has used the health club facilities?

what is prob. that an employee chosen at random is a male?

what is the prob. that an employee chosen at random is a male or has not used the health club facilities?

Solution

Here,

2×2 contingency table prepared from the given information which is as follows:

		Health Club Facilities		Total
		male	female	
Health Club Facilities	Yes	65	? $(110 - 65) = 45$	110
	No	? $(170 - 65) = 105$	? $(80 - 45) = 35$	? $(250 - 110) = 140$
Total		170	? $(250 - 170) = 80$	250

Prob. that an employee chosen at random is a female and has used the health club facilities = $\frac{45}{250} = 0.18$

Prob that an employee chosen at random is a male = $\frac{170}{250} = 0.68$

Let $P(A)$ = prob. of choosing a male = $\frac{170}{250} = 0.68$

$P(B)$ = prob. that he/she not used health club facilities

$$= \frac{140}{250} = 0.56$$

$P(AB) = \text{prob. that an employee is a male and has not used the health club facilities} = \frac{105}{250} = 0.42$

Now,

Prob. that an employee is a male and has not used the health club facilities $= \frac{105}{250}$
 $= 0.42$

Now,

prob. that an employee chosen at random is a male or has not used the health club facilities

$$P(A \cup B) = P(A) + P(B) - P(AB)$$

$$= 0.68 + 0.56 - 0.42$$

$$= 0.82$$



8. The prob. that a management trainee will remain with the Company is 0.6 and the prob. ~~with~~ that the trainee will earn more than RS 50,000 per year is 0.2. The prob. that an employee is a management trainee who remained with the Company or who earned more than RS 5,000 per year is 0.5. What is the prob. that an employee earned more than 5000 per year given that he remained with the Company?

Solⁿ given that ~~दिया की है~~ ^{है} $\therefore$ Conditional prob. है,

Here,

Let A: Event of trainee remain with Company

B: Event of trainee earns more than RS 5000

Then, $P(A) = 0.60$

$P(B) = 0.2$

$$P(A \cup B) = 0.50$$

$$P(B/A) = ?$$

We know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{or, } 0.50 = 0.60 + 0.20 - P(A \cap B)$$

$$\text{or, } P(A \cap B) = 0.80 - 0.50 \Rightarrow 0.30$$

Now,

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{0.30}{0.60}$$

$$= 0.50$$



Bayes Theorem

It is propounded by the British mathematician Thomas Bayes in 1763.

It is the extended version of Conditional Probability.

If there are two or more mutually exclusive events to intersect a common event, then the question is related with Bayes theorem.

[दुई वा तीनवटा अलग तर्फे जाबक - फल, event मिलेर हुने सब common event सिर्जनाओ अवस्थामा ती Bayes Theorem ले]

Example:

mutually exclusive events:

common events

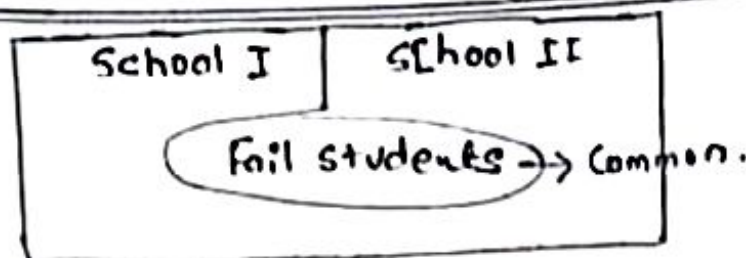
(a) machine X, Y and Z

defective product

(b) city A, B & C

Smokers.

If we have two mutually exclusive events



Exclusive Events $P(E_1) =$	Common Events $P(F E_1) = \dots$	Product (i.e. multiply) $P(E_1) \times P(F E_1) = \dots$ Let (Bharat)
$P(E_2) =$	$P(F E_2)$	$P(E_2) \times P(F E_2) = \dots$ (Let pokharel)
Sum = 1		Sum = $P(F) = P(\text{fail students})$ Bharat + Pokharel = Let ABC

Now,

$$P(E_1|F) = \text{prob. of selecting college 1 for fail students} \\ = \frac{\text{Bharat}}{\text{ABC}}$$

$$P(E_2|F) = \text{prob. of selecting college 2 for fail students.} \\ = \frac{\text{pokharel}}{\text{ABC}}$$

If we have three mutually exclusive events

Exclusive Events	Common Events	Product (i.e. multiply)
$P(M_1)$	$P(D M_1) = \dots$	$P(M_1) \times P(D M_1) = \dots$
$P(M_2)$	$P(D M_2) = \dots$	$P(M_2) \times P(D M_2) = \dots$
$P(M_3)$	$P(D M_3) = \dots$	$P(M_3) \times P(D M_3) = \dots$
Sum = 1		Sum = $\dots$

Exclusive Events $P(P_1) = 0.60$	Common Events $P(S/P_1) = 0.01$	Product $0.60 \times 0.01 = 0.006$
$P(P_2) = 0.25$	$P(S/P_2) = 0.02$	$0.25 \times 0.02 = 0.005$
$P(P_3) = 0.15$	$P(S/P_3) = 0.02$	$0.15 \times 0.02 = 0.003$
Sum = 1		Sum = $P(S) = 0.014$

Req. (a)

Prob. of getting Sub-standard quality is given by $P(S) = 0.014$

Req. (b). In order to find from which plant the Sub-standard product is expected to be manufactured, we need to calculate following probabilities.

$$P(P_1/S) = \frac{0.006}{0.014} = 0.428$$

$$P(P_2/S) = \frac{0.005}{0.014} = 0.357$$

$$P(P_3/S) = \frac{0.003}{0.014} = 0.215$$

Decision: Referring above calculations, from plant 1, the Sub-standard product is expected to be manufactured as it has a higher prob. of 0.428. ✓

Bayes' Theorem

माना कि तीन निम्न संघट्ट 1

$$P(V) = P(A_1) \cdot P(V/A_1) + P(A_2) \cdot P(V/A_2) + P(A_3) \cdot P(V/A_3)$$

$$P\left(\frac{A_1}{V}\right) = \frac{P(A_1) \cdot P(V/A_1)}{P(A_1) \cdot P(V/A_1) + P(A_2) \cdot P(V/A_2) + P(A_3) \cdot P(V/A_3)}$$

$$P(A_1) \cdot P(V/A_1) + P(A_2) \cdot P(V/A_2) + P(A_3) \cdot P(V/A_3)$$

$$P\left(\frac{A_2}{V}\right) = \frac{P(A_2) \cdot P(V/A_2)}{P(A_1) \cdot P(V/A_1) + P(A_2) \cdot P(V/A_2) + P(A_3) \cdot P(V/A_3)}$$

$$P\left(\frac{A_3}{V}\right) = \frac{P(A_3) \cdot P(V/A_3)}{P(A_1) \cdot P(V/A_1) + P(A_2) \cdot P(V/A_2) + P(A_3) \cdot P(V/A_3)}$$

A factory produces electric bulbs by three machines. The machine A, B and C produces 3000, 2500 and 4500 bulbs per day respectively. According to past experience, it is known that the defective bulbs produced by three machines it is known that the defective bulbs produced by three machines are 2%, 1% & 3% respectively.

- ① A bulb is selected at random from a day's total production, find the prob. that it is defective. (ii) A bulb is drawn at random from a day's total production and is defective, find the prob. that it was produced by (a) machine A, (b) machine B, (c) machine C.

Sol

Here,

Bulbs produced by machine A	= 3000
" " " "	(B) = 2500
" " " "	(C) = 4500
Total bulbs	= 10,000



∴ Prob. that the bulbs produced by machine A.

$$P(A) = \frac{3000}{10000} = \frac{3}{10} = 0.3$$

∴ Prob. that the bulbs produced by machine B.

$$P(B) = \frac{2500}{10000} = 0.25$$

∴ Prob. that the bulbs produced by machine C.

$$P(C) = \frac{4500}{10000} = 0.45$$

Let D be the event that the produced bulbs are defective

$$\therefore P(D/A) = 2\% = 0.02$$

$$\therefore P(D/B) = 3\% = 0.03$$

$$\therefore P(D/C) = 4\% = 0.04$$



The prob. that the selected

(i) defective prob.

$$= P(A) \times P(D/A) + P(B) \times P(D/B) + P(C) \times P(D/C)$$

$$= 0.3 \times 0.02 + 0.25 \times 0.03 + 0.45 \times 0.04$$

$$= 0.0345$$

The prob. that the selected defective bulbs is produced from machine A is

$$\begin{aligned} P(A/D) &= \frac{P(A) \times P(D/A)}{P(A) \times P(D/A) + P(B) \times P(D/B) + P(C) \times P(D/C)} \\ &= \frac{0.03 \times 0.02}{0.0315} \\ &= 0.1905 \end{aligned}$$


Again, The prob. that the selected defective bulbs is produced from machine B is

$$\begin{aligned} P(B/D) &= \frac{P(B) \times P(D/B)}{P(A) \times P(D/A) + P(B) \times P(D/B) + P(C) \times P(D/C)} \\ &= \frac{0.25 \times 0.03}{0.0315} \Rightarrow 0.238 \end{aligned}$$

Again, The prob. that the selected defective bulbs is produced from machine C is

$$\begin{aligned} P(C/D) &= \frac{P(C) \times P(D/C)}{P(A) \times P(D/A) + P(B) \times P(D/B) + P(C) \times P(D/C)} \\ &= \frac{9/20 \times 0.04}{0.0315} \\ &= 0.571 \end{aligned}$$

(9) The chances of X, Y, Z becoming manager of a certain company are 4, 2, 3. The prob. that bonus scheme will be introduced if X, Y, Z becomes manager are 0.3, 0.5 & 0.8 respectively. If the bonus scheme has been introduced, what is the prob. that X is appointed as the manager?

Sol?

Here,

Let A_1, A_2 and A_3 denotes the event of becoming a manager for X, Y & Z respectively.

$$P(A_1) = \frac{4}{4+2+3} = \frac{4}{9} = 0.44$$

$$P(A_2) = \frac{2}{4+2+3} = \frac{2}{9} = 0.22$$

$$P(A_3) = \frac{3}{4+2+3} = \frac{3}{9} = 0.33$$

Let B denotes the events for bonus scheme

$$P(B/A_1) = 0.3$$

$$P(B/A_2) = 0.5$$

$$P(B/A_3) = 0.8$$

Now, the prob. that X is appointed as the manager if the bonus scheme has been introduced is computed as

$$P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + P(A_3) \cdot P(B|A_3)}$$

$$= \frac{0.132}{0.132 + 0.11 + 0.264}$$

$$= 0.26$$

Hence, the required prob. is 26%.



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